

Cross-Chip Gates by Coherent State Transfer Only: Ping-Pong vs. Migration (No Mid-Circuit Measurement)

Companion to `locc_ejpp_tutorial.tex`; figures from `make_state_transfer_circuits.py`

July 2026

Abstract

The EJPP telegate implements a cross-chip gate with one ebit and two mid-circuit measurements. This note works out the fully *coherent* alternative: use the cable’s quantum state transfer (QST) as a unitary “move” operation and do every two-qubit gate locally. No measurement, no feedforward, no classical bits, no Pauli frame. We give the transfer operator, prove the two implementation patterns (*ping-pong*: two cable traversals per gate; *migration*: two traversals *total*), show the exact algebraic condition that decides which one a circuit admits, and apply both to `HF_8q_3doubles_rzx.json`. All circuits shown here were verified unitarily equivalent to their nonlocal targets by the companion script.

1 The transfer operator

From `qst_formulas.txt`, the cable executes a full state transfer in $\tau \approx 66$ ns:

$$(\alpha|0\rangle + \beta|1\rangle)_A |0\rangle_R |0\rangle_B \longrightarrow |0\rangle_A |0\rangle_R (\alpha|0\rangle + \beta|1\rangle)_B,$$

i.e. on the (sender, receiver) pair, with the receiver guaranteed to start in $|0\rangle$, it acts as a SWAP. We write T for the transfer sender→receiver and note two properties used throughout:

1. **Self-cleaning $|0\rangle$ bookkeeping.** T leaves the sender in $|0\rangle$ — which is exactly the empty receiver that the return transfer $T^\dagger$ requires. The “hole” shuttles back and forth inside the communication pair and never touches data.
2. **Disjoint support $\Rightarrow$ commutation.** T acts only on the travelling qubit, the cable mode, and the ancilla. For any gate V on *other* qubits,

$$T V T^\dagger = V \quad (\text{equivalently } T V = V T).$$

2 Toy model

Four qubits. Chip A holds data qubits A_0, A_1 (A_1 is the seam qubit that will travel); chip B holds data qubit B_0 and a communication ancilla e initialized to $|0\rangle$. The target circuit is three cross-chip gates with local layers in between:

$$U = R_3 V_2 R_2 V_1 R_1, \quad R_k = R_{ZZ}(t_k) \text{ on } (A_1, B_0), \quad (1)$$

with $V_1 = H_{A_0} \otimes R_Z(\varphi)_{A_1}$ and $V_2 = H_{A_0} \otimes Z_{B_0}$. Note V_1 touches the travelling qubit A_1 , but only with a *single-qubit* gate — this matters below.

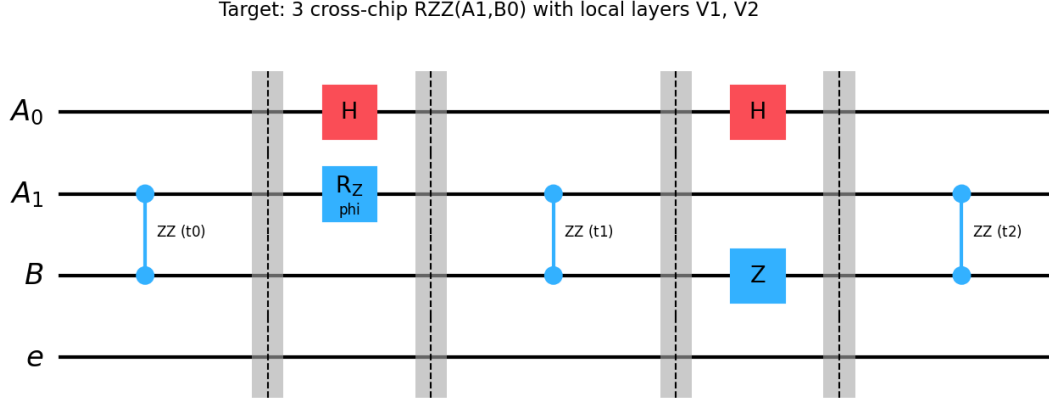


Figure 1: The nonlocal target (1): three $R_{ZZ}(t_k)$ straddle the chip cut (between A_1 and B_0); the ancilla e is idle. This is the circuit we must reproduce without any gate crossing the cut.

2.1 Ping-pong: two traversals per gate

The basic identity: a nonlocal R_{ZZ} equals transfer out, local gate against the ancilla, transfer back,

$$R_{ZZ}(\theta)_{A_1 B_0} = T^\dagger R_{ZZ}(\theta)_{e B_0} T. \quad (2)$$

Proof on a basis state: T carries the bit value x of A_1 onto e ; the local gate imprints $e^{-i\frac{\theta}{2}(-1)^{x \oplus y}}$, exactly the phase the nonlocal gate would have imprinted; $T^\dagger$ returns x to A_1 . Substituting (2) for each R_k in (1):

$$U = (T^\dagger L_3 T) V_2 (T^\dagger L_2 T) V_1 (T^\dagger L_1 T), \quad L_k \equiv R_{ZZ}(t_k)_{e B_0}. \quad (3)$$

Six T 's appear: **six cable traversals, with the data on the wire every time** (Fig. 2). Everything is unitary and deterministic — already no measurement is needed.

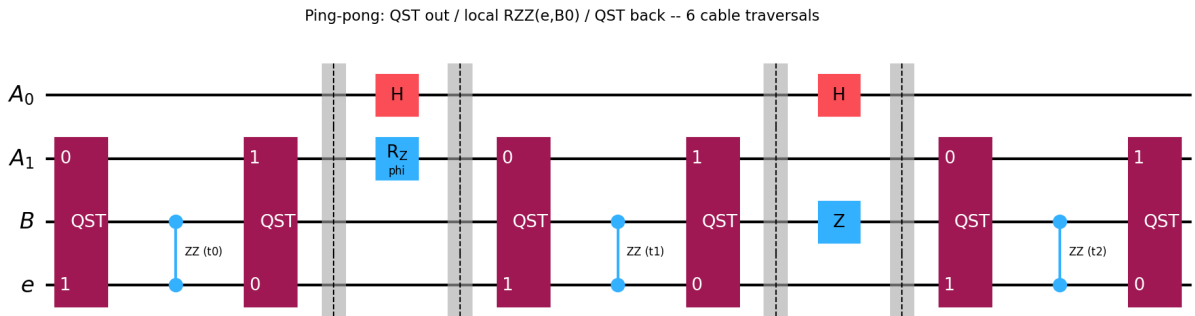


Figure 2: Ping-pong implementation of Fig. 1. Each QST box is one cable traversal (drawn as the exact SWAP it enacts on the required subspace). Every R_{ZZ} is now local on chip B; cost: 6 traversals.

2.2 Migration: two traversals total

If every intermediate layer commutes with T , the inner transfers cancel. Slide each V_i through the neighbouring transfers in (3) using $TV_iT^\dagger = V_i$:

$$U = T^\dagger L_3 \underbrace{(TV_2T^\dagger)}_{=V_2} L_2 \underbrace{(TV_1T^\dagger)}_{=V'_1} L_1 T = T^\dagger (L_3 V_2 L_2 V'_1 L_1) T. \quad (4)$$

All adjacent $T^\dagger T$ pairs annihilate. Read right to left: transfer out *once*, run the entire dressed sequence with the state living on chip B, transfer home *once* (Fig. 3).

Two remarks on the condition:

- **Single-qubit gates on the traveller are free.** V_1 contains $R_Z(\varphi)$ on A_1 ; conjugation by T just relabels it onto e : $V'_1 = H_{A_0} \otimes R_Z(\varphi)_e$. The gate travels with the qubit and stays local.
- **Chip-B gates on the traveller are also free** — they were the point of migrating.

The only forbidden pattern is stated in Section 2.3.

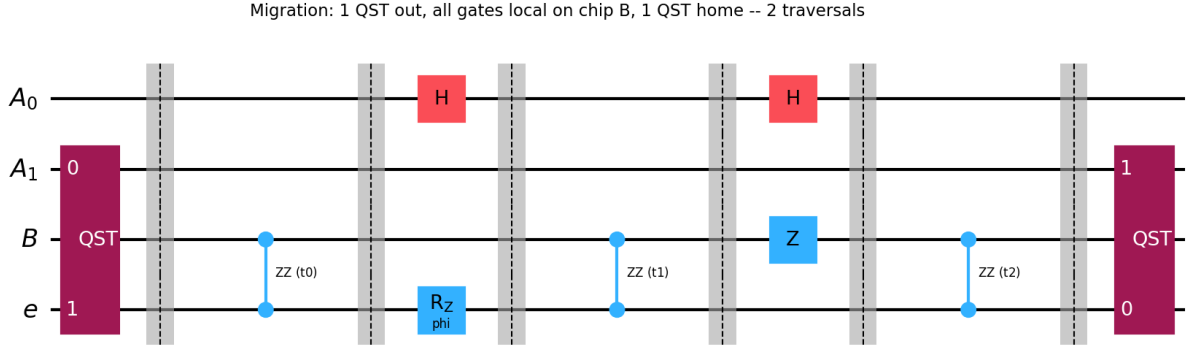


Figure 3: Migration implementation of the same target: one QST out, all three $R_{ZZ}(t_k)$ and the $R_Z(\varphi)$ dressing local on chip B, one QST home. Cost: 2 traversals. Verified unitarily identical to Fig. 1.

2.3 When migration is blocked

Let $V_1 = CZ(A_0, A_1)$: a two-qubit gate between the traveller and a qubit that *stayed on chip A*. Conjugation by T relabels $A_1 \rightarrow e$:

$$T CZ_{A_0 A_1} T^\dagger = CZ_{A_0 e},$$

which spans the cut — itself a cross-chip gate (Fig. 4). The telescoping in (4) dies at this layer: either bring the qubit home (back to ping-pong) or perform a nonlocal CZ (the thing being avoided). Hence the rule:

Migration condition. While the seam qubit is abroad, it may receive any single-qubit gate and any two-qubit gate with destination-chip qubits, but *no two-qubit gate with a qubit of its home chip*.

Blocked: $V_1 = \text{CZ}(A_0, A_1)$ commutes into $\text{CZ}(A_0, e)$, itself cross-chip

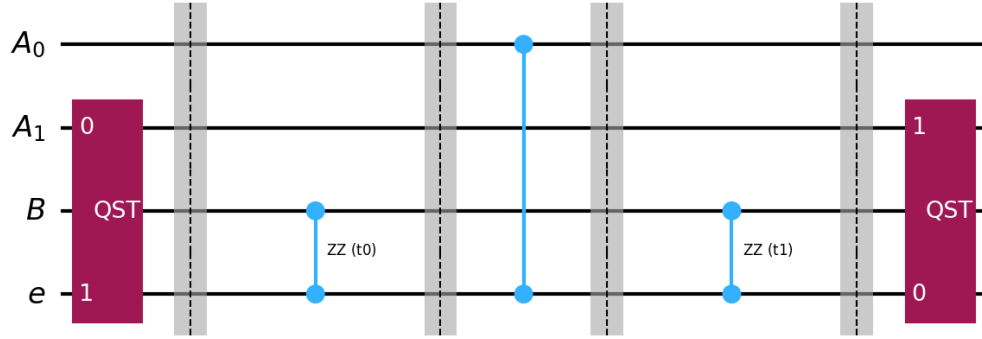


Figure 4: The counterexample: with $V_1 = \text{CZ}(A_0, A_1)$, migrating A_1 turns the layer into $\text{CZ}(A_0, e)$ — the gate drawn here between the top wire (chip A) and the bottom wire (chip B) crosses the cut, so the 2-traversal scheme fails.

3 Application to HF_8q_3doubles_rzx

Chip A = $q_0 \dots q_3$, chip B = $q_4 \dots q_7$ plus one communication ancilla e . The only cross-chip gates are the three $R_{ZX}(t_k)$ on (q_6, q_2) .

3.1 What runs today: ping-pong (exact)

Replace each $R_{ZX}(t_k)_{q_6 q_2}$ by $T_{q_2 \rightarrow e} R_{ZX}(t_k)_{q_6 e} T_{e \rightarrow q_2}$. Figure 5 shows the full 9-wire circuit; the companion script verifies it equals the original 8-qubit unitary. Cost: 6 traversals, zero measurements, total added seam time $\approx 6 \times 66 \text{ ns} \approx 0.4 \mu\text{s}$ (plus local routing), negligible against $T_1 = 85 \mu\text{s}$.

HF_8q_3doubles_rzx_ping-pong QST version (exact; chip A=q0-q3, chip B=q4-q7,e)

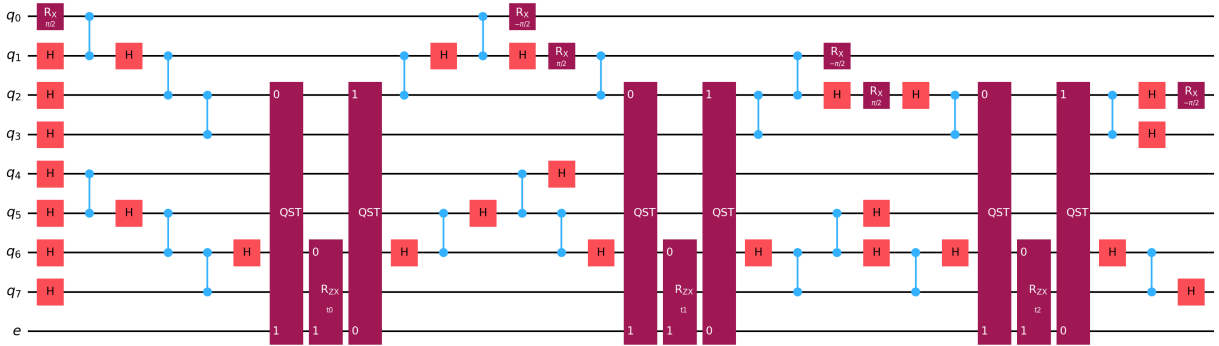


Figure 5: HF_8q_3doubles_rzx in ping-pong form: exact, runnable with the current ansatz. Each crossing is QST out / local R_{ZX} / QST back.

3.2 Why this ansatz cannot migrate

Between $R_{ZX}(t_0)$ and $R_{ZX}(t_1)$ the circuit applies $CZ(q_1, q_2)$ (with gates on q_1 in between, so the pair does not cancel) — with q_2 abroad this is exactly the blocked pattern of Section 2.3. The mirror choice fails identically: q_6 participates in $CZ(q_5, q_6)$ between crossings, and q_5 stays on chip B. The parity ladders of the three doubles re-dress both seam qubits between every pair of crossings.

3.3 The desired circuit structure

Figure 6 is the *template* a redesigned ansatz must match: q_2 migrates once; between crossings, chip A layers V_A act only on $\{q_0, q_1, q_3\}$ (the q_2 wire is empty), while chip B layers V_B may freely dress $q_4 \dots q_7$ and even e (single-qubit gates on the migrated state are allowed); after the last crossing q_2 returns. Total: 2 traversals for all three doubles.

Concretely, the redesign task for the excitation ladders is: route the fermionic parity of each double onto q_6 using *chip-B* CNOT/CZ ladders, and onto the α -sector representative *before* the migration (or after the return), so that q_2 's only two-qubit partners while abroad are q_6 /chip-B qubits. Whether the three doubles $(3, 7 \rightarrow 4, 0)$, $(3, 7 \rightarrow 5, 1)$, $(3, 7 \rightarrow 6, 2)$ admit such an ordering at chemical accuracy is checkable classically by exact diagonalization before anything is run.

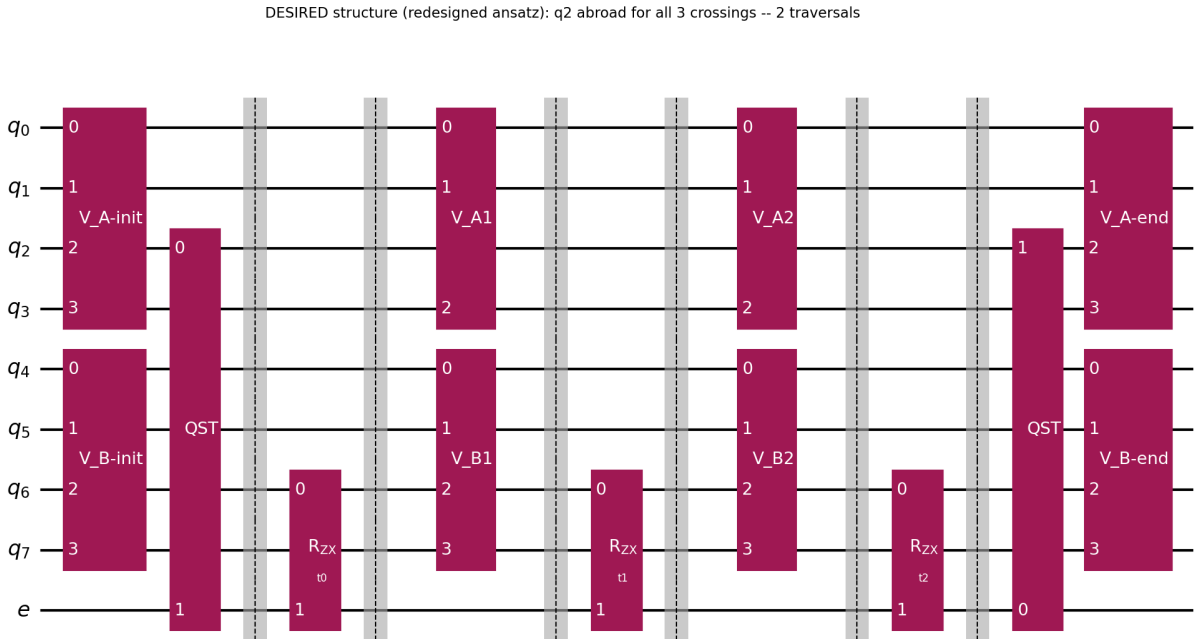


Figure 6: Desired structure for a redesigned ansatz (schematic; V boxes are placeholders): q_2 lives on chip B for the whole doubles block, all three $R_{ZX}(t_k)$ are local, and the cable is used exactly twice.

4 Cost comparison

Per-use cable noise p (depolarizing), measurement 500+200 ns, feedback 200 ns, QST 66 ns, local 2q gate 50 ns:

Scheme	Cable uses on data	Measurements	Seam time
Direct $R_{ZX} \times 3$	3 (2-qubit exposure)	0	~ 150 ns
Ping-pong (Fig. 5)	6 (1-qubit exposure)	0	$\sim 0.5 \mu s$
Migration (Fig. 6)	2 (1-qubit exposure)	0	$\sim 0.3 \mu s$
EJPP telegate $\times 3$	0 (ancilla-only)	6	$\sim 5-6 \mu s$

At equal per-use noise, ping-pong roughly ties the direct gate ($6 \times p/2$ vs $3 \times \frac{15}{16}p$ average infidelity) — its value is that it wins whenever a passive 66 ns transfer is cleaner than a full cross-chip entangling gate, and that it needs no measurement hardware. Migration is a strict $3\times$ improvement over ping-pong and beats the direct gate at any p , but requires the ansatz redesign above. EJPP keeps data off the cable entirely (and allows heralding/distillation) at the price of mid-circuit measurement, readout error, and feedforward latency. All coherent schemes still require the seam-located twirling+PEC layer to reach chemical accuracy at $p \sim 0.1$.

Reproducibility

All figures and all unitary-equivalence checks (`Operator.equiv` at random parameter bindings) are produced by `make_state_transfer_circuits.py` in this folder:

```
python3 make_state_transfer_circuits.py
```